

DocuMind RAG Stress-Test Corpus

Synthetic enterprise dossier designed to test semantic retrieval, chunking, metadata filtering, OCR, grounding, numerical reasoning, temporal consistency, multi-hop questions, and hallucination resistance.

Corpus use: upload this PDF as a single document into DocuMind. Then run the evaluation questions from the separate Answer Key PDF. The corpus intentionally contains distractors, near-duplicate facts, tables, negations, dates, and an image-only appendix.

Document ID for human reference: DST-2026-09. All names, teams, organizations, and data in this corpus are fictional.

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1. Northstar Operations Overview

Northstar Research Institute operates three internal products: Orion, Atlas, and Beacon. Orion is the primary document intelligence platform. Atlas is the structured reporting platform. Beacon is the internal notification service.

Orion is designed for document upload, extraction, semantic retrieval, and grounded question answering. Standard documents are retained for 180 days after their last modification unless a retention exception applies.

Retention exception: documents tagged LEGAL-HOLD do not follow the 180-day deletion rule. They remain retained until the legal-hold record is explicitly released.

Northstar time standard: all audit timestamps are stored in UTC. User-facing dashboards may render local time, but the database source of truth is UTC.

Support hours: Monday-Friday, 09:00-18:00 India Standard Time, excluding company holidays. Critical incidents are handled 24x7.

The Orion service has two search modes. Global search considers all documents owned by the authenticated user. Document-scoped search considers only the selected document. Both modes enforce ownership filtering.

A document can have status PROCESSING, READY, or FAILED. PROCESSING means ingestion has started; READY means text/chunks/embeddings completed successfully; FAILED means ingestion stopped because processing could not produce a usable result.

A completed answer should be grounded in retrieved document context. The assistant should not invent a fact merely because the question sounds plausible.

2. Orion Platform Specification

Orion version 4.2 uses a 384-dimensional embedding model for the retrieval prototype. Vector search is performed in PostgreSQL using pgvector. Cosine distance is used for the current retrieval implementation; smaller distance indicates greater similarity.

The current prototype uses Qwen3 0.6B through a local LM Studio server for answer generation. The model is not the source of truth for user documents; retrieved document chunks are the source of truth.

Chunking guideline for the current prototype: target approximately 500 characters per chunk. The production chunker may later become token-aware and structure-aware.

Retrieval defaults: top-k is 3 in the chat API, while direct search allows the caller to specify a limit. A document_id filter may be supplied to restrict retrieval to a single document.

Security rule: document_id is never sufficient by itself. Every document retrieval must also be constrained by the authenticated user_id.

Example vector distances from a benchmark run:
Query: "What is my father name?"
Chunk A: 0.693
Chunk B: 0.978
Chunk C: 1.064
For this implementation, Chunk A ranks ahead of B and C because its distance is smallest.

The embedding service generates one query vector for one user question. During document ingestion, it generates one vector per stored chunk.

3. Support and Access Policies

Support severity is defined as follows:

Severity	Definition	Target response
S1	Service unavailable for all users or suspected data-loss event	15 minutes
S2	Major feature unavailable for a subset of users	1 hour
S3	Non-critical defect with workaround	1 business day
S4	Cosmetic issue or documentation defect	3 business days

Access roles: Reader can search owned documents. Editor can upload, update metadata, and delete owned documents. Admin can perform organization-wide maintenance. A Reader does not automatically gain access to other users' documents.

Password rule: passwords are stored only as Argon2 hashes. The API never returns password hashes in a user response schema.

Deletion rule: deleting a document record should also delete its corresponding local storage file. If the file is missing, the deletion workflow should still remove the metadata row but log the storage inconsistency.

Unsupported upload types: executable files are not accepted. Supported prototype types include PDF, DOCX, TXT, JPG, JPEG, and PNG. File extension is treated as a filter, while actual file content is checked before processing.

OCR rule: an image-based page with little or no extractable text should be rendered and passed through OCR. Mixed pages may contain both embedded text and images.

4. Deployment Matrix

The following table is authoritative for deployment ownership in this corpus.

Service	Owner team	Primary environment	Backup environment	RTO
Orion API	Platform Engineering	Production-East	Production-West	30 min
Orion Worker	Platform Engineering	Production-East	Production-West	45 min
Atlas Reports	Data Systems	Production-West	Production-East	2 hr
Beacon Alerts	Site Reliability	Production-East	Production-West	15 min

The Orion API and Orion Worker are separate services even though both are owned by Platform Engineering. The API accepts HTTP requests; the Worker is responsible for asynchronous document processing in the planned architecture.

Current prototype note: document processing is still synchronous in the API request path. The worker architecture described above is planned, not yet implemented in the current prototype.

For incident response, the service owner must be contacted before changing production configuration. Changing the backup environment without a recorded incident ticket is prohibited.

5. Incident Reports and Root Causes

Incident INC-401 - Vector extension failure
Date: 2026-09-11. Symptom: an Alembic migration failed with a PostgreSQL error stating that type vector did not exist. Root cause: the Python pgvector package was installed, but the PostgreSQL server extension itself was not installed and enabled. Resolution: install the PostgreSQL pgvector extension, run CREATE EXTENSION vector in the target database, then rerun the migration.

Incident INC-402 - Truncated PNG upload
Date: 2026-09-11. Symptom: OCR returned no text and Pillow raised "image file is truncated" when the saved PNG was loaded. Evidence: uploaded size and saved size did not match. Resolution: reset the UploadFile pointer, read the complete bytes, write those bytes to storage, and verify the saved file size equals the uploaded byte count.

Incident INC-403 - Empty extraction falsely marked READY
Date: 2026-09-11. Symptom: some image PDFs completed processing but produced zero document_chunks. Root cause: the pipeline allowed an empty extracted string to reach the chunker, which returned no chunks. Resolution: treat empty extracted text as processing failure and do not mark the document READY.

Incident INC-404 - LLM answered "I don't know" even when the correct chunk was retrieved
Date: 2026-09-16. Symptom: retrieval returned a chunk containing the answer, but Qwen3 0.6B responded that it did not know. Root cause: too much context was supplied to a very small model. Resolution: reduce the retrieval limit for the test to 1, use a more explicit grounded prompt, set temperature to 0, and keep the answer short.

Incident INC-405 - Oversized chunk reduced retrieval quality
Date: 2026-09-16. Symptom: a 1,410-character chunk mixed a tree paragraph, personal information, and unrelated text. Root cause: the prototype chunker was too large and the embedding model max sequence length was 256 tokens. Resolution: reduce the prototype chunk target to about 500 characters and plan a future token-aware chunker.

Important: these incidents are intentionally included so the evaluation questions can test whether the assistant distinguishes between symptoms, root causes, and fixes rather than simply matching keywords.

6. People and Roles

Mira Sen is Director of Platform Engineering. She approves production architecture changes for Orion.

Dev Malik is the Orion API lead. He owns API routing, authentication integration, and response schema changes.

Leena Rao is the Orion retrieval lead. She owns embedding evaluation, chunking experiments, and vector search quality.

Arjun Mehta is the Orion ingestion lead. He owns PDF extraction, OCR, file validation, and document processing reliability.

Sana Iyer is the incident coordinator. She tracks incident tickets and change records.

A role is not the same as a permission. A person may own a component without automatically receiving organization-wide administrative access.

For the benchmark, the following names are intentionally similar enough to test entity resolution: Mira Sen and Sana Iyer both coordinate cross-team work, but only Mira approves production architecture changes; Sana tracks incident records.

7. Finance and Capacity Tables

Quarterly operating capacity for the Orion ingestion worker:

Quarter	Documents processed	Average pages/doc	OCR share	Failure rate
Q1 2026	4,800	6.2	18%	2.5%
Q2 2026	5,400	6.8	21%	2.1%
Q3 2026	6,300	7.1	27%	1.9%
Q4 2026	6,900	7.6	31%	1.8%

Q3 2026 budget allocation for the prototype: engineering 46%, storage 18%, observability 14%, model infrastructure 12%, security review 10%.

A separate planning note states that model infrastructure may rise to 16% in a future quarter, but the Q3 2026 budget above remains authoritative for Q3.

Arithmetic note: percentage categories in a single budget row sum to 100%. Do not assume that a larger OCR share means a larger absolute failure count without checking document volume.

8. Change Log and Superseded Values

The following records show how a value can change over time. When answering a question that specifies a date, use the value valid for that date.

Date	Item	Value	Status
2026-06-01	Orion default top-k	5	Superseded
2026-07-15	Orion default top-k	4	Superseded
2026-08-20	Orion default top-k	3	Current
2026-07-01	Embedding model dimension	384	Current
2026-08-01	Chat context limit	3 chunks	Current

A June 2026 design note says the prototype planned to use synchronous processing. A September 2026 design note says a background worker is planned but not yet implemented. If asked what is implemented now, answer that synchronous processing is still used in the current prototype.

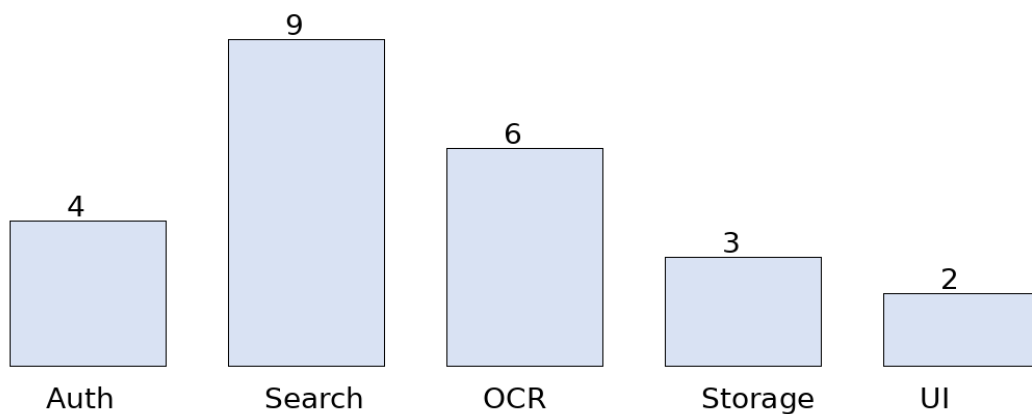
A policy note from an earlier draft mentioned a 90-day retention period. That draft is superseded by the current 180-day retention rule in Section 1.

9. Mixed Evidence Page

The following paragraph is real embedded PDF text. The chart below is an image object and must be read from the visual evidence if OCR is functioning.

The Q3 incident review identified Search as the subsystem with the largest number of reported defects in the visual chart. The chart also shows that Storage had fewer defects than OCR.

Quarterly defect counts by subsystem



Visual values are part of the evidence; OCR should recover them.

If the chart and the paragraph appear to disagree, inspect the underlying values. The image values are: Auth 4, Search 9, OCR 6, Storage 3, UI 2.

10. OCR-only Field Inspection Appendix

The page image below is intentionally inserted as an image-only page. It contains no embedded text layer. Use this page to test the OCR path of your document processor.

OCR Appendix - Field Inspection Card

Inspection ID: OCR-17

Site: Harbor Annex

Inspector: Mina Kapoor

Signal quality: 87 percent

Required action: replace filter before 18 October 2026

Do not restart the pump until the red lockout tag is removed.

Recorded temperature: 42.6 C

Priority: MEDIUM

This page is intentionally image-only for OCR testing.

11. Retrieval Trap Notes

The following statements are intentionally close in wording but have different meanings:

1. The retention period is 180 days for ordinary documents.
2. A LEGAL-HOLD document is not deleted after 180 days unless the hold is released.
3. The 90-day retention period appears only in an earlier superseded draft.
4. The current prototype uses a 3-chunk chat retrieval default, but the direct search endpoint allows a caller-specified limit.
5. The current prototype does not yet use a background worker even though one is planned.

Use these distinctions to test negation, temporal reasoning, and "current vs planned" questions.